How do I remember variance va precision

Here's a tiny cheat-sheet to keep **variance** vs **precision** straight.

Variance vs Precision (remember this!)

- **Definitions**
  - Variance: σ^2 (spread).
  - Standard deviation: $\sigma = \sqrt{\sigma^2}$.
  - **Precision**: $\tau = \dfrac{1}{\sigma^2}$ (inverse variance).
- **Units**
  - Variance has units of "thing^2".
  - Precision has units of "1/thing^2".
- **Gaussian notation**
  - $x \sim \mathcal{N}(\mu, \sigma^2) \;=\; \mathcal{N}\left(\mu, \dfrac{1}{\tau}\right).$
- **Bayesian linear regression (Bishop/PRML convention)**
  - Prior on weights: $\mathbf{w} \sim \mathcal{N}\big(\mathbf{0},\ \alpha^{-1}\mathbf{I}\big)$
    ⇒ **α = prior precision**, prior variance per weight $= 1/\alpha$.
  - Likelihood: $y \mid \mathbf{w} \sim \mathcal{N}\big(\Phi\mathbf{w},\ \beta^{-1}\mathbf{I}\big)$
    ⇒ **β = noise precision**, noise variance $= 1/\beta$.
- **Convert quickly**
  - Given std s: $\sigma^2 = s^2,\ \ \tau = 1/s^2$.
  - Given variance v: $\tau = 1/v,\ \ s = \sqrt{v}$.
  - Given precision p: $v = 1/p,\ \ s = 1/\sqrt{p}$.
- **Numerical mini-examples**
  - "std = 1/5": $s = 0.2 \Rightarrow \sigma^2 = 0.04 \Rightarrow \tau = 25.$
  - "prior precision = 2": $\tau = 2 \Rightarrow \sigma^2 = 1/2 \Rightarrow s = \sqrt{1/2} \approx 0.7071.$
- **Mnemonics**
  - **V is for Volume (spread); P is for Pinpoint (peaky)**.
  - Bigger **precision** ⇒ peak is **pointier** (smaller variance).
  - See α, β in formulas? Think **precisions**. See σ^2? That's **variance**.
- **Common pitfalls**

- Don't "invert twice." If code expects **variance**, pass $1/\beta$ (not β).
  If it expects **precision**, pass β (not $1/\beta$).
- In multivariate normals, libraries usually want **covariance** Σ.
  If you have precision Λ, use $\Sigma = \Lambda^{-1}$.
- In Laplace/MAP, the **Hessian ≈ posterior precision**; the **covariance is its inverse**.

If you want, paste the exact API you're calling (JAX/NumPy/Scipy function) and I'll mark **exactly** whether it wants std, variance, covariance, or precision.